Experiment 3: Antenna Characterization

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1 Aim

To determine the operating frequency and impedance matching of an antenna via VSWR measurements, and to characterize the given antenna by measuring its beam pattern in both co-polarization and cross-polarization configurations.

2 Objectives

1. To determine the antenna operating frequency, reflection coefficient, and return loss from VSWR measurements using the RF Tee-junction and oscilloscope method, as well as direct instrument measurement.
2. To measure the effect of distance on the power received by the antenna and verify the inverse-square propagation trend.
3. To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations across the E-plane and H-plane.
4. To determine the half-power beamwidth (HPBW) in both planes and estimate the directivity of the antenna.

3 Apparatus Required

- Monopole and dipole antennas
- Horn antenna / Log-Periodic Antenna under test (DUT)
- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyzer / Vector Network Analyzer (VNA)
- RF Tee connector
- 50 Ω termination load
- Antenna support stand with graduated rotation turntable
- Coaxial cables and suitable RF adapters

4 Theory

4.1 Basic Antenna Concepts

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. An antenna can be used either as a transmitting antenna or as a receiving antenna.

A dipole antenna has two equal metal arms in a straight line and is fed at the centre. Its total length is approximately half a wavelength ($\lambda/2$), with a feed line connected between the two arms:

$$L_{\text{dipole}} \approx \frac{\lambda}{2} \tag{1}$$

It radiates most strongly in directions perpendicular to the wire and has minimal radiation along the wire axis.

A monopole antenna is approximately half of a dipole mounted perpendicularly above a conducting ground plane. Its length is approximately one-quarter wavelength ($\lambda/4$):

$$L_{\text{monopole}} \approx \frac{\lambda}{4} \implies \lambda = 4L_{\text{monopole}} \tag{2}$$

The ground acts like a mirror through the method of electromagnetic images, making it behave like a full dipole in the upper half-space. The operating frequency f relates to the speed of light c (3×10^8 m/s) via:

$$f = \frac{c}{\lambda} \tag{3}$$

4.2 Reflection Coefficient, VSWR, and Return Loss

When the antenna impedance Z_L differs from the characteristic impedance Z_0 of the transmission line, a part of the incident signal is reflected. The voltage reflection coefficient at the antenna terminals is:

$$\Gamma = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{4}$$

The magnitude $|\Gamma|$ indicates the degree of impedance mismatch. For a perfectly matched antenna ($Z_L = Z_0$), $|\Gamma| = 0$.

In the RF Tee-junction setup, the oscilloscope monitors the standing-wave ripple resulting from interference between the incident and reflected waves. Measuring the maximum and minimum peak-to-peak voltages yields the Voltage Standing Wave Ratio (VSWR):

$$VSWR = \frac{V_{\max}}{V_{\min}} \tag{5}$$

The magnitude of the reflection coefficient is related to VSWR by:

$$|\Gamma| = \frac{VSWR - 1}{VSWR + 1} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{6}$$

The return loss (RL) in decibels is defined as:

$$RL = -20 \log_{10} |\Gamma| \quad \text{dB} \tag{7}$$

A well-matched antenna exhibits a VSWR close to unity ($VSWR \leq 2$ represents an acceptable operational bandwidth), a small reflection coefficient, and a high return loss.

4.3 Antenna Beam Pattern and Principal Planes

The energy radiated by an antenna is represented by its radiation pattern:

- **Main Lobe:** The region where the maximum radiated energy exists, pointing along the principal boresight direction.
- **Side Lobes:** Minor lobes where radiation spreads sideways.
- **Back Lobe:** The radiation lobe oriented 180° opposite to the main lobe.

E-plane and H-plane angular measurements characterize the radiation beam pattern of an antenna:

- **E-Plane:** The plane containing the electric-field vector and the propagation vector.
- **H-Plane:** The plane containing the magnetic-field vector and the propagation vector. The E-plane and H-plane are mutually perpendicular.

4.4 Polarization Configurations

- **Co-Polarization:** The electric fields of both transmitting and receiving antennas are aligned parallel in the same plane, measuring maximum power reception.
- **Cross-Polarization:** The transmitting antenna's electric field is oriented strictly perpendicular to that of the receiving antenna (90° offset). This measures polarization impurity, leakage, and cross-polarization discrimination (XPD).

4.5 Half-Power Beamwidth (HPBW) and Directivity

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two points where the received power drops to half of its maximum value, corresponding to the -3 dB points relative to the peak normalized pattern:

$$P_{\text{norm}}(\theta)\ [\text{dB}] = P_r(\theta)\ [\text{dBm}] - P_{\max}\ [\text{dBm}] \tag{8}$$

Directivity (D) quantifies the concentration of radiation in the principal direction compared with an isotropic radiator:

$$D = \frac{4\pi}{\Omega_A} \approx \frac{4\pi}{\theta_E \theta_H} \tag{9}$$

where θ_E and θ_H are the half-power beamwidths of the E-plane and H-plane expressed in radians.

4.6 Distance Dependence and Power Conversion

By the Friis transmission equation, the received power in the far field follows an inverse-square dependence:

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 \propto \frac{1}{R^2} \tag{10}$$

Expressed logarithmically:

$$P_r\ (\text{dBm}) = C - 20\log_{10}(R) \tag{11}$$

which exhibits a linear decline of -1 dB/dB against $20\log_{10}(R)$. Power measured in dBm is converted to linear milliwatts (mW) via:

$$P_r\ (\text{mW}) = 10^{\frac{P_r\ (\text{dBm})}{10}} \tag{12}$$

5 Procedure

5.1 Part A: VSWR and Operating Frequency Determination

1. For the test monopole antenna, design the rod length for a chosen resonant frequency of 300 MHz using the quarter-wavelength relation ($L = \lambda/4$). Adjust the monopole length to 25 cm.

2. Connect the RF signal generator, an RF Tee connector, the antenna under test (DUT), and the digital storage oscilloscope (set to 1 MΩ input impedance) as shown in the standard VSWR measurement setup.

3. Keep the generator output amplitude constant and perform frequency sweeps around 300 MHz and an off-resonance frequency (120 MHz).

4. Record the maximum peak-to-peak voltage ($V_{\max}$) and minimum peak-to-peak voltage ($V_{\min}$) on the oscilloscope screen.

5. Calculate the VSWR, reflection coefficient ($|\Gamma|$), and return loss (RL) for both frequencies.

6. For the horn antenna, connect the feed terminal directly to the calibrated RF analyzer/VNA instrument to capture the broadband VSWR curve.

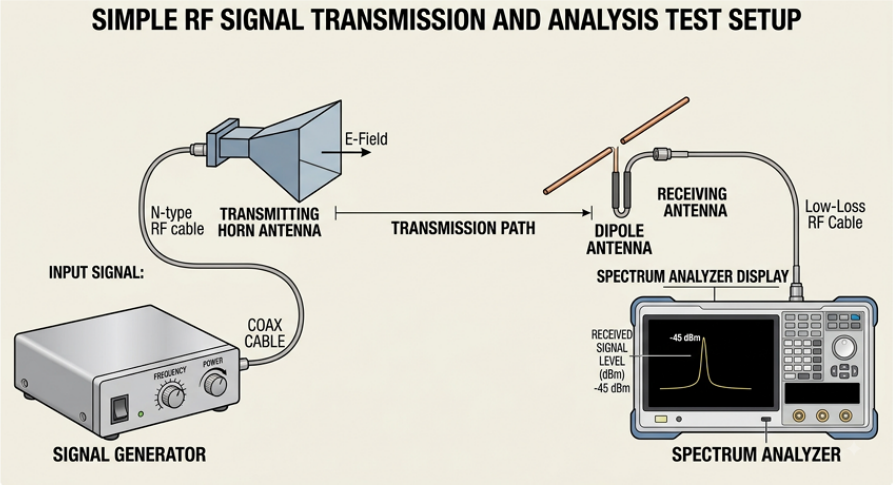


Figure 1: Experimental setup

5.2 Part B: Resonant Dimensioning for Radiation Measurements

1. Measure the exposed physical length of the internal monopole feed probe inside the horn antenna.

2. Compute the theoretical quarter-wave resonant frequency and evaluate the error relative to the actual cutoff frequency of 1.42 GHz.

3. Adjust the transmitting dipole antenna length to half the wavelength ($\lambda/2$) corresponding to 1.42 GHz.

5.3 Part C: Effect of Distance

1. Align the transmitting dipole and receiving horn at the same height with parallel polarization along their line of sight.
2. Set the RF signal generator to 1.42 GHz at a constant output power.
3. Systematically vary the distance R between the antennas from 2.40 m to 8.20 m.
4. Record the received power in dBm using the spectrum analyzer.
5. Convert the received power from dBm to linear milliwatts (mW).
6. Plot P_r (mW) vs. R and P_r (dBm) vs. R.

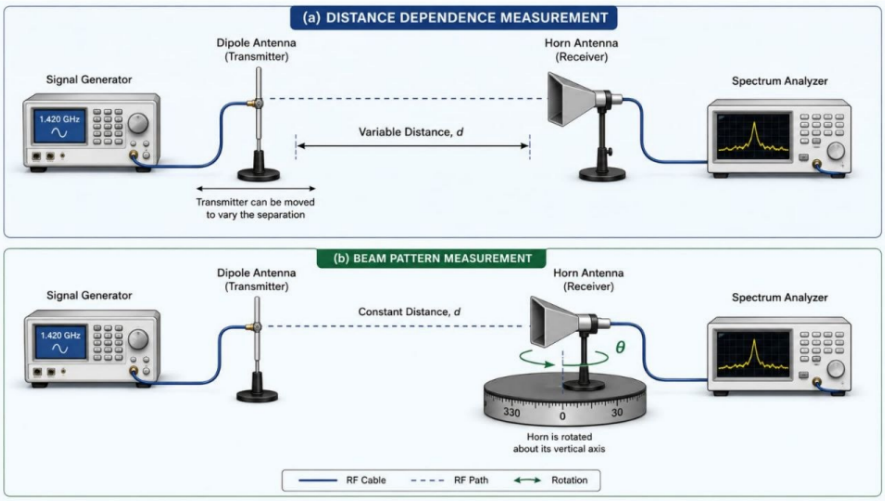


Figure 2: Experimental setup for distance-dependance and beaam-pattern measurement.

5.4 Part D: Beam Pattern – Co-Polarization Measurement

1. Position the receiving antenna on a rotatable stand at a separation distance of 420 cm (4.20 m).
2. Align the electric fields of the transmitter and receiver in the same plane (co-polarization).
3. Starting at 0° boresight, rotate the antenna through 360° in steps of 10°.
4. Record the received power for both H-plane and E-plane co-polarization sweeps.
5. Normalize the power profiles, plot Cartesian and polar radiation patterns, and identify the −3 dB HPBW points.

5.5 Part E: Beam Pattern – Cross-Polarization Measurement

1. Rotate the transmitting antenna feed by 90° such that its electric field is perpendicular to the receiving antenna's electric field.
2. Maintain the identical separation distance (420 cm), frequency (1.42 GHz), and input power.

3. Rotate the receiving antenna through a full circle and record the received power in both E-plane and H-plane cross-polarization orientations.

4. Plot the cross-polarization patterns and compare the peak levels against the co-polarization main beam to evaluate cross-polarization rejection.

6 Observations

6.1 VSWR Measurement Data (Monopole Antenna)

- Target design frequency: $f_0 = 300$ MHz
- Theoretical wavelength: $\lambda = c/f_0 = (3 \times 10^8)/(300 \times 10^6) = 1.0$ m $= 100$ cm
- Monopole length set: $L = \lambda/4 = 25$ cm

Table 1: Oscilloscope Standing Wave Measurements for Monopole Antenna

Frequency	$V_{\mathbf{max}}$ **(p-p)**	$V_{\mathbf{min}}$ **(p-p)**	**VSWR**	$\lvert\Gamma\rvert$	**Return Loss (dB)**
300 MHz	6×500 mV $= 3.0$ V	4×500 mV $= 2.0$ V	1.50	0.200	13.98 dB
120 MHz	4.5×1.0 V $= 4.5$ V	0.8×500 mV $= 0.4$ V	11.25	0.837	1.55 dB

6.2 Distance vs. Received Power Data

Table 2: Received Power Measurements with Varying Distance at $f = 1.42$ GHz

S. No.	**Distance** R **(cm)**	**Distance** R **(m)**	**Received Power** P_r **(dBm)**	**Linear Power** P_r **(mW**
1	240	2.40	-34.65	3.428×10^{-4}
2	346	3.46	-38.26	1.493×10^{-4}
3	470	4.70	-44.81	3.304×10^{-5}
4	596	5.96	-42.87	5.164×10^{-5}
5	638	6.38	-43.50	4.467×10^{-5}
6	820	8.20	-53.00	5.012×10^{-6}

6.3 Co-Polarization Radiation Pattern Data ($R = 420$ cm)

Table 3: Co-Polarization Angular Measurements for H-Plane and E-Plane at $R = 420$ cm

Angle (°)	**H-Plane Co-Pol (dBm)**	**E-Plane Co-Pol (dBm)**
0	-43.13	-58.50
10	-42.81	-61.30
20	-42.60	-68.70
30	-43.33	-66.20
40	-45.01	-64.20
50	-46.39	-63.60
60	-47.26	-61.40

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Angle (°)	H-Plane Co-Pol (dBm)	E-Plane Co-Pol (dBm)
70	-48.71	-61.20
80	-49.89	-70.10
90	-49.89	-68.30
100	-57.07	-62.10
110	-59.63	-60.50
120	-67.71	-59.70
130	-53.41	-60.70
140	-55.92	-59.30
150	-52.86	-59.10
160	-51.46	-59.00
170	-53.22	-60.50
180	-56.74	-57.70
190	-54.04	-56.60
200	-53.09	-57.00
210	-55.86	-60.00
220	-59.03	-68.20
230	-52.23	-66.40
240	-52.31	-62.20
250	-50.53	-61.10
260	-56.21	-61.20
270	-54.32	-60.50
280	-59.44	-75.30
290	-59.48	-64.90
300	-52.96	-59.10
310	-52.41	-56.20
320	-54.62	-55.90
330	-52.95	-56.90
340	-48.07	-57.00
350	-45.89	-57.10
360	-44.32	-58.70

6.4 Cross-Polarization Radiation Pattern Data ($R = 420$ cm)

Table 4: Cross-Polarization Angular Measurements for E-Plane and H-Plane at $R = 420$ cm

Angle (°)	E-Plane Cross-Pol (dBm)	H-Plane Cross-Pol (dBm)
0	-36.73	-51.50
10	-36.79	-52.20
20	-37.65	-51.00
30	-39.09	-51.20
40	-40.77	-50.80
50	-41.96	-50.10
60	-41.16	-50.00
70	-45.97	-49.00

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Angle (°)	E-Plane Cross-Pol (dBm)	H-Plane Cross-Pol (dBm)
80	-51.75	-48.09
90	-62.16	-52.90
100	-54.53	-57.90
110	-50.00	-60.20
120	-50.90	-63.10
130	-53.80	-61.10
140	-52.60	-59.30
150	-49.77	-63.70
160	-46.33	-62.30
170	-45.44	-58.40
180	-44.71	-60.20
190	-44.71	-63.20
200	-48.88	-58.40
210	-52.38	-60.20
220	-54.92	-63.20
230	-54.92	-67.20
240	-56.47	-63.40
250	-52.67	-59.10
260	-51.63	-56.40
270	-52.09	-56.10
280	-63.35	-55.20
290	-66.65	-55.40
300	-46.45	-55.20
310	-43.13	-54.20
320	-41.41	-51.60
330	-40.97	-50.50
340	-40.97	-50.20
350	-39.74	-48.20
360	-36.72	-51.60

6.5 Summary of Results

Table 5: Antenna Characterization Summary

Quantity	Symbol	Value
Calculated Monopole Frequency (Horn probe)	f_{calc}	1.875 GHz
Actual Horn Cutoff Frequency	f_{actual}	1.420 GHz
Percentage Error in Frequency	$\Delta f/f$	32.04%
Transmitting Dipole Length	L_{dipole}	10.56 cm
VSWR at 300 MHz (Tuned Monopole)	$VSWR_{300}$	1.50 ($RL = 13.98$ dB)
VSWR at 120 MHz (Untuned Monopole)	$VSWR_{120}$	11.25 ($RL = 1.55$ dB)
H-Plane Half-Power Beamwidth	θ_H	53.23° (0.929 rad)
E-Plane Half-Power Beamwidth	θ_E	60.74° (1.060 rad)
Estimated Directivity (Linear)	D	12.76
Estimated Directivity (Decibels)	D_{dBi}	11.06 dBi

7 Calculations

7.1 VSWR, Reflection Coefficient, and Return Loss Calculations

For the designed 300 MHz quarter-wave monopole antenna ($L = 25$ cm):

1. At Resonant Frequency ($f = 300$ MHz): Oscilloscope readings:

$$V_{\max} = 6 \times 500 \text{ mV} = 3000 \text{ mV} = 3.0 \text{ V}$$
$$V_{\min} = 4 \times 500 \text{ mV} = 2000 \text{ mV} = 2.0 \text{ V}$$

The Voltage Standing Wave Ratio is:

$$VSWR = \frac{V_{\max}}{V_{\min}} = \frac{3.0 \text{ V}}{2.0 \text{ V}} = \mathbf{1.50} \tag{13}$$

The magnitude of the reflection coefficient is:

$$|\Gamma| = \frac{VSWR - 1}{VSWR + 1} = \frac{1.50 - 1}{1.50 + 1} = \frac{0.50}{2.50} = \mathbf{0.200} \tag{14}$$

The return loss in decibels is:

$$RL = -20 \log_{10}(|\Gamma|) = -20 \log_{10}(0.200) = -20(-0.6990) = \mathbf{13.98} \text{ dB} \tag{15}$$

Because $VSWR = 1.50 \leq 2.0$ and $RL \approx 14$ dB, the antenna is well-matched at its intended resonant operating frequency.

2. At Off-Resonant Frequency ($f = 120$ MHz): Oscilloscope readings:

$$V_{\max} = 4.5 \times 1.0 \text{ V} = 4.5 \text{ V}$$
$$V_{\min} = 0.8 \times 500 \text{ mV} = 0.4 \text{ V}$$

The Voltage Standing Wave Ratio is:

$$VSWR = \frac{V_{\max}}{V_{\min}} = \frac{4.5 \text{ V}}{0.4 \text{ V}} = \mathbf{11.25} \tag{16}$$

The magnitude of the reflection coefficient is:

$$|\Gamma| = \frac{11.25 - 1}{11.25 + 1} = \frac{10.25}{12.25} \approx \mathbf{0.8367} \tag{17}$$

The return loss in decibels is:

$$RL = -20 \log_{10}(0.8367) = -20(-0.0774) = \mathbf{1.55} \text{ dB} \tag{18}$$

The high VSWR ($11.25 \gg 2$) and low return loss confirm severe impedance mismatch away from resonance, reflecting over 70% of incident power back to the source.

7.2 Horn Antenna Cutoff Frequency and Error Analysis

Using the measured visible length of the monopole feed probe inside the horn ($L_{\text{meas}} = 4.0$ cm $= 0.04$ m):

$$\lambda_{\text{calc}} = 4 \times L_{\text{meas}} = 4 \times 0.04 \text{ m} = 0.16 \text{ m} = 16 \text{ cm}$$

$$f_{\text{calc}} = \frac{c}{\lambda_{\text{calc}}} = \frac{3 \times 10^8 \text{ m/s}}{0.16 \text{ m}} = 1.875 \times 10^9 \text{ Hz} = 1.875 \text{ GHz}$$

Comparing this with the actual calibrated cutoff operating frequency ($f_{\text{actual}} = 1.42$ GHz):

$$\text{Absolute Error } (\Delta f) = |1.875 - 1.42| \text{ GHz} = 0.455 \text{ GHz}$$

$$\text{Percentage Error} = \frac{|1.875 - 1.42|}{1.42} \times 100\% = \frac{0.455}{1.42} \times 100\% \approx \mathbf{32.04}\%$$

Physical Explanation of Error: The exposed probe length of 4.0 cm only accounts for the wire segment protruding into the horn waveguide cavity. The internal connector transition consists of a recessed center pin, mounting sleeve, and dielectric insulator (PTFE/Teflon) seated inside the metallic cavity wall. This internal path contributes to the effective electrical length. In addition, open-ended monopoles exhibit capacitive end-fringing, and the metallic waveguide boundary adds reactive loading, reducing the resonant frequency from 1.875 GHz down to 1.42 GHz.

For an ideal quarter-wavelength at 1.42 GHz:

$$L_{\text{ideal}} = \frac{c}{4 f_{\text{actual}}} = \frac{3 \times 10^8}{4 \times 1.42 \times 10^9} \text{ m} \approx 5.28 \text{ cm} \tag{19}$$

The difference of $5.28 \text{ cm} - 4.00 \text{ cm} \approx 1.28$ cm represents the combined contribution of internal connector housing, dielectric loading, and capacitive fringing.

7.3 Transmitting Dipole Resonant Length

To radiate efficiently at $f = 1.42$ GHz, the wavelength is:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{1.42 \times 10^9 \text{ Hz}} = 0.21126 \text{ m} \approx 21.13 \text{ cm} \tag{20}$$

Because a center-fed dipole operates at $\lambda/2$:

$$L_{\text{dipole}} = \frac{\lambda}{2} = \frac{21.126 \text{ cm}}{2} \approx \mathbf{10.56} \text{ cm} \tag{21}$$

Each arm of the transmitting dipole is therefore set to 5.28 cm.

7.4 Power Conversion from dBm to Linear Milliwatts (mW)

Using $P_{\text{mW}} = 10^{(P_{\text{dBm}}/10)}$:

- $R = 2.40$ m: $P_r = -34.65$ dBm $\implies P_{\text{mW}} = 10^{-3.465} = 3.428 \times 10^{-4}$ mW
- $R = 3.46$ m: $P_r = -38.26$ dBm $\implies P_{\text{mW}} = 10^{-3.826} = 1.493 \times 10^{-4}$ mW
- $R = 4.70$ m: $P_r = -44.81$ dBm $\implies P_{\text{mW}} = 10^{-4.481} = 3.304 \times 10^{-5}$ mW
- $R = 5.96$ m: $P_r = -42.87$ dBm $\implies P_{\text{mW}} = 10^{-4.287} = 5.164 \times 10^{-5}$ mW
- $R = 6.38$ m: $P_r = -43.50$ dBm $\implies P_{\text{mW}} = 10^{-4.350} = 4.467 \times 10^{-5}$ mW
- $R = 8.20$ m: $P_r = -53.00$ dBm $\implies P_{\text{mW}} = 10^{-5.300} = 5.012 \times 10^{-6}$ mW

7.5 Half-Power Beamwidth (HPBW) Calculations

1. H-Plane Co-Polarization HPBW (θ_H):

- Peak power: $P_{\max} = -42.60$ dBm at $\theta = 20°$.
- -3 dB Threshold: $-42.60 - 3.00 = -45.60$ dBm (or -3.00 dB normalized).
- Right crossing: Interpolating between $40°$ (-45.01 dBm) and $50°$ (-46.39 dBm) yields $\theta_1 = 44.28°$.
- Left crossing: Interpolating between $360°$ (-44.32 dBm) and $350°$ (-45.89 dBm) yields $\theta_2 = 351.05°$ ($-8.95°$).
- $\theta_H = 44.28° - (-8.95°) = \mathbf{53.23}° = \mathbf{0.929}$ rad.

2. E-Plane Co-Polarization HPBW (θ_E):

- Peak power: $P_{\max} = -55.90$ dBm at $\theta = 320°$.
- -3 dB Threshold: $-55.90 - 3.00 = -58.90$ dBm (or -3.00 dB normalized).
- Left crossing: Interpolating between $300°$ (-59.10 dBm) and $310°$ (-56.20 dBm) yields $\theta_1 = 300.69°$.
- Right crossing: Interpolating between $0°$ (-58.50 dBm) and $10°$ (-61.30 dBm) yields $\theta_2 = 1.43°$ ($\equiv 361.43°$).
- $\theta_E = (360° - 300.69°) + 1.43° = \mathbf{60.74}° = \mathbf{1.060}$ rad.

7.6 Antenna Directivity Estimation

Using the half-power beamwidths in radians:

$$D \approx \frac{4\pi}{\theta_E \theta_H} = \frac{4\pi}{(1.060 \text{ rad})(0.929 \text{ rad})} = \frac{12.5664}{0.9847} \approx \mathbf{12.76} \tag{22}$$

In decibels relative to an isotropic radiator:

$$D \text{ (dBi)} = 10 \log_{10}(12.76) \approx \mathbf{11.06} \text{ dBi} \tag{23}$$

8 Graphical Plots

9 Precautions

1. Maintain identical heights and optical line-of-sight alignment between transmitting and receiving antennas.
2. Keep the RF signal generator frequency and output power strictly constant during sweeps.
3. For oscilloscope Tee-junction measurements, set channel input impedance to 1 MΩ to avoid loading the transmission line.
4. Keep the antenna propagation path free from reflecting objects, laboratory furniture, and personnel to minimize multipath interference.
5. Avoid sharp bending of RF coaxial cables to prevent impedance discontinuities and phase distortion.

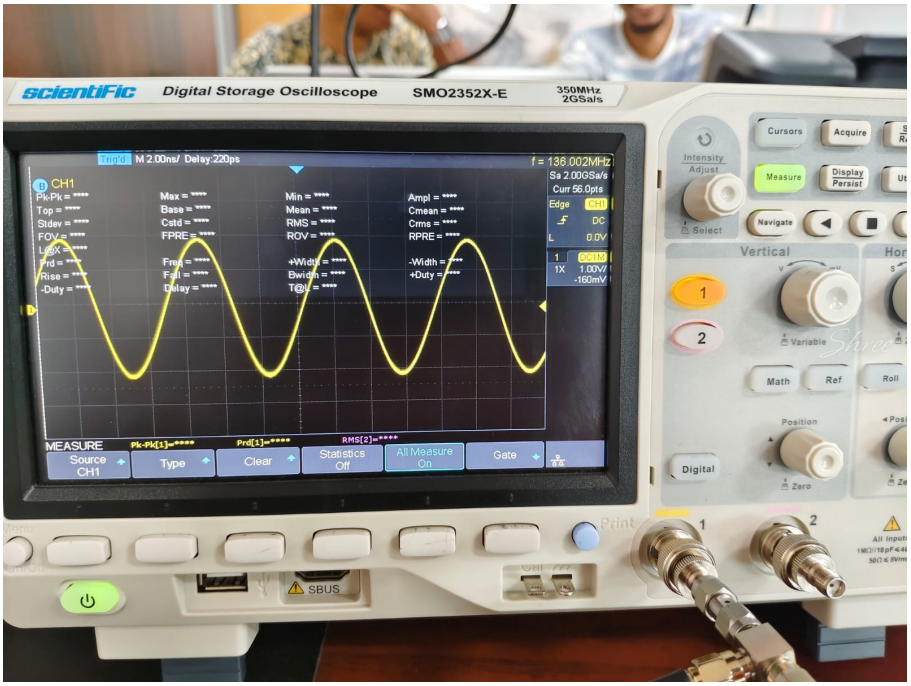


Figure 3: Direct instrument screen capture displaying the VSWR vs. Frequency response of the horn antenna.

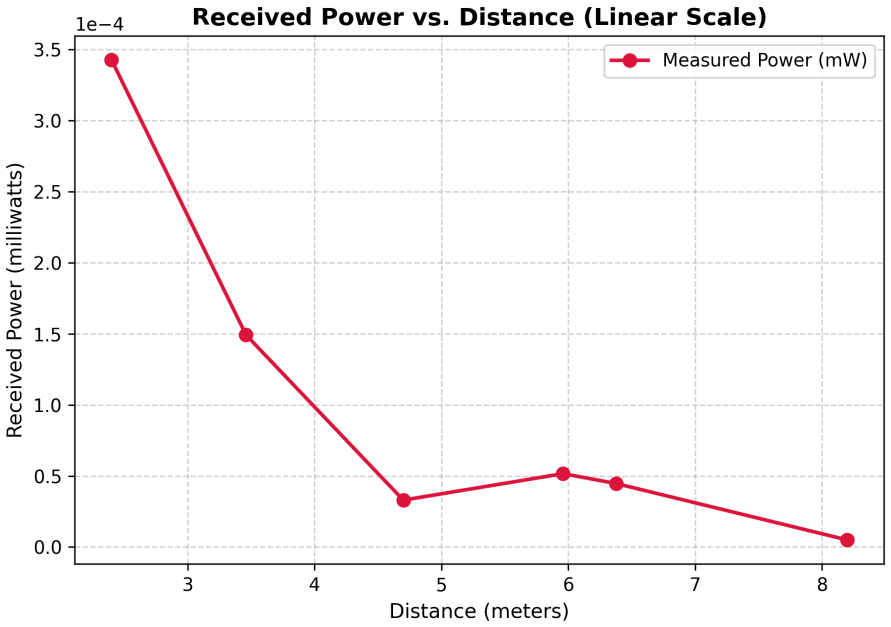


Figure 4: Received Power (P_r in mW) vs. Distance (R in m) exhibiting $1/R^2$ decay.

10 Conclusion

1. For the designed 300 MHz quarter-wave monopole ($L = 25$ cm), the measured standing wave pattern gave $VSWR = 1.50$ and Return Loss $= 13.98$ dB, verifying resonance and good impedance matching ($VSWR \leq 2$). Off resonance at 120 MHz, $VSWR$ degraded to 11.25 ($RL = 1.55$ dB).

2. For the horn antenna, broadband VSWR was verified directly via the RF network analyzer

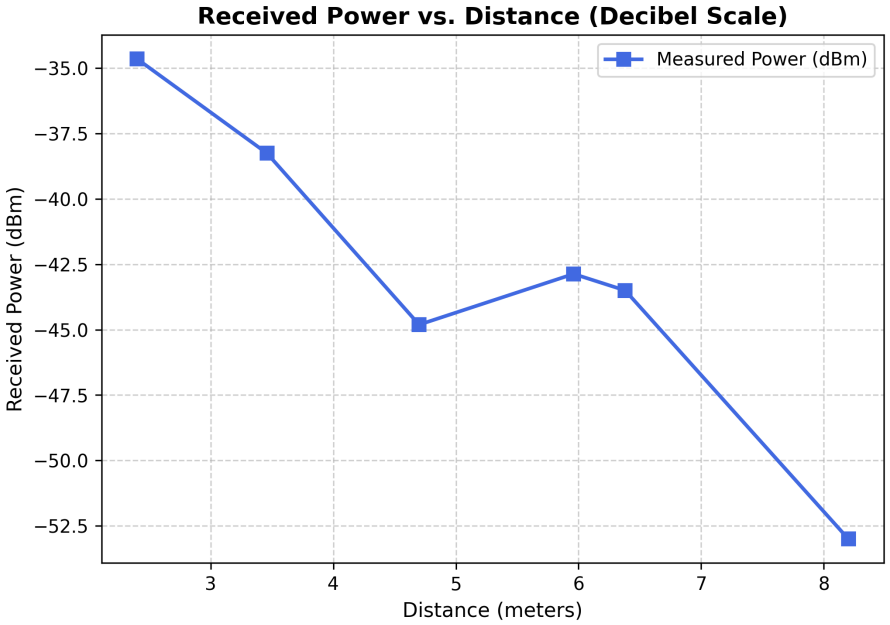


Figure 5: Received Power (P_r in dBm) vs. Distance (R in m) illustrating logarithmic path loss.

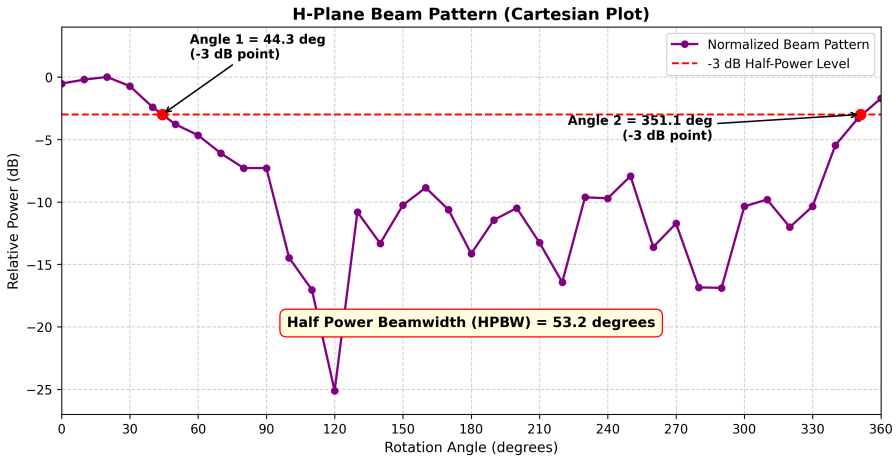


Figure 6: H-Plane Co-Polarization Pattern at $R = 420$ cm in Cartesian coordinates with -3 dB HPBW points.

instrument, and its feed cutoff frequency was measured as 1.42 GHz. The exposed probe length gave an apparent resonance of 1.875 GHz (32.04% error), explained by the recessed connector transition, dielectric loading, and capacitive end-fringing.

3. The transmitting dipole was configured to $\lambda/2 = 10.56$ cm to match the horn's 1.42 GHz cutoff frequency.

4. Received power vs. distance verified the inverse-square law ($1/R^2$) in linear units and logarithmic path-loss behavior in decibels.

5. Half-power beamwidths were measured as $\theta_H = 53.23°$ and $\theta_E = 60.74°$, yielding an estimated directivity of $D \approx 12.76$ (11.06 dBi). Cross-polarization patterns showed significant rejection of orthogonal electric fields, demonstrating polarization selectivity.

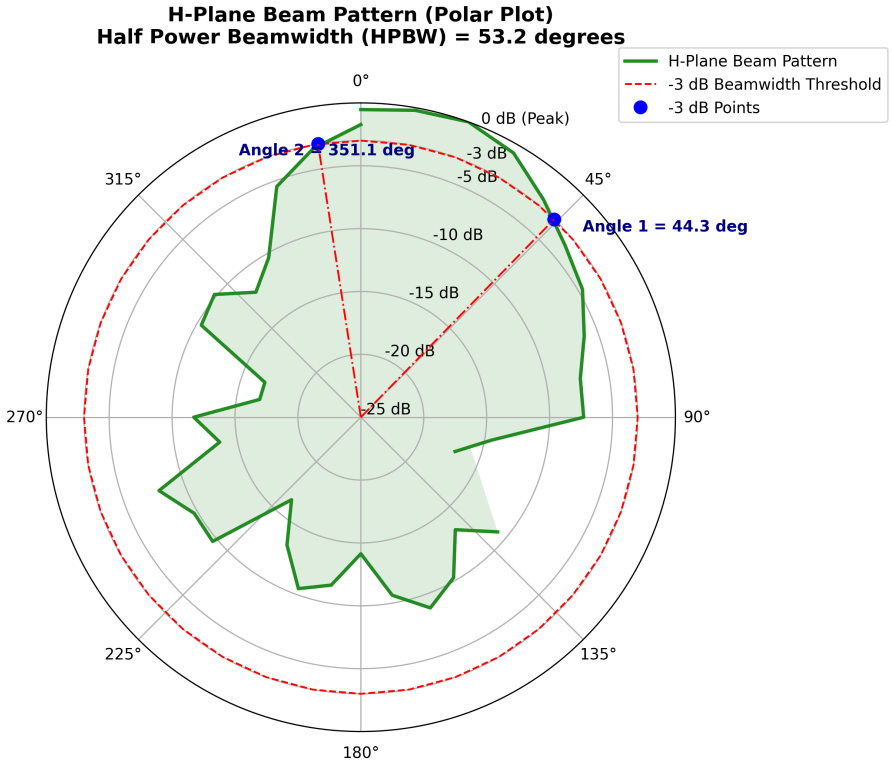


Figure 7: H-Plane Co-Polarization Pattern at $R = 420$ cm in Polar coordinates ($\Delta\theta_H = 53.23°$).

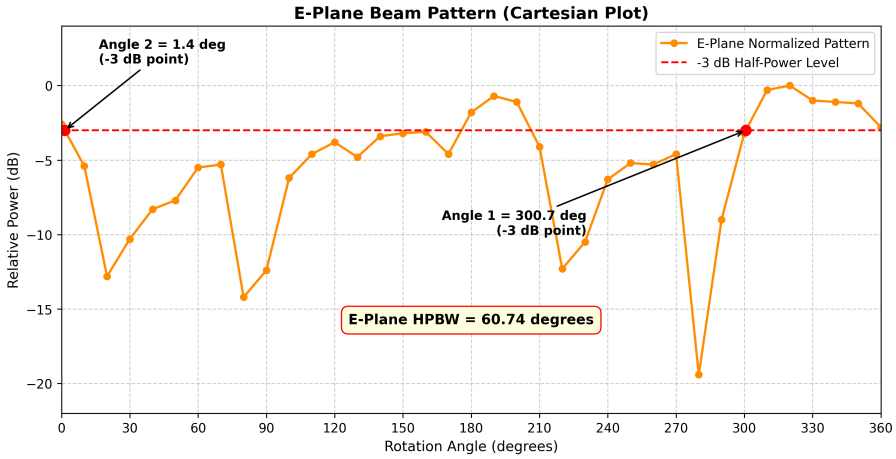


Figure 8: E-Plane Co-Polarization Pattern at $R = 420$ cm in Cartesian coordinates with -3 dB HPBW points.

11 Discussion Questions

Q1. Why does the plot of P_r (dBm) vs. R look visually similar to P_r (dBm) vs. $20\log_{10}(R)$?

Power in dBm is already a logarithmic quantity (P_r (dBm) $\propto -20\log_{10}(R)$). Over the experimental distance range from 2.40 m to 8.20 m (a factor of 3.42, which is well under one decade), the curvature of $\log_{10}(R)$ is modest. Consequently, both curves appear nearly straight. In contrast, converting to linear power in milliwatts reveals the steep $1/R^2$ decay.

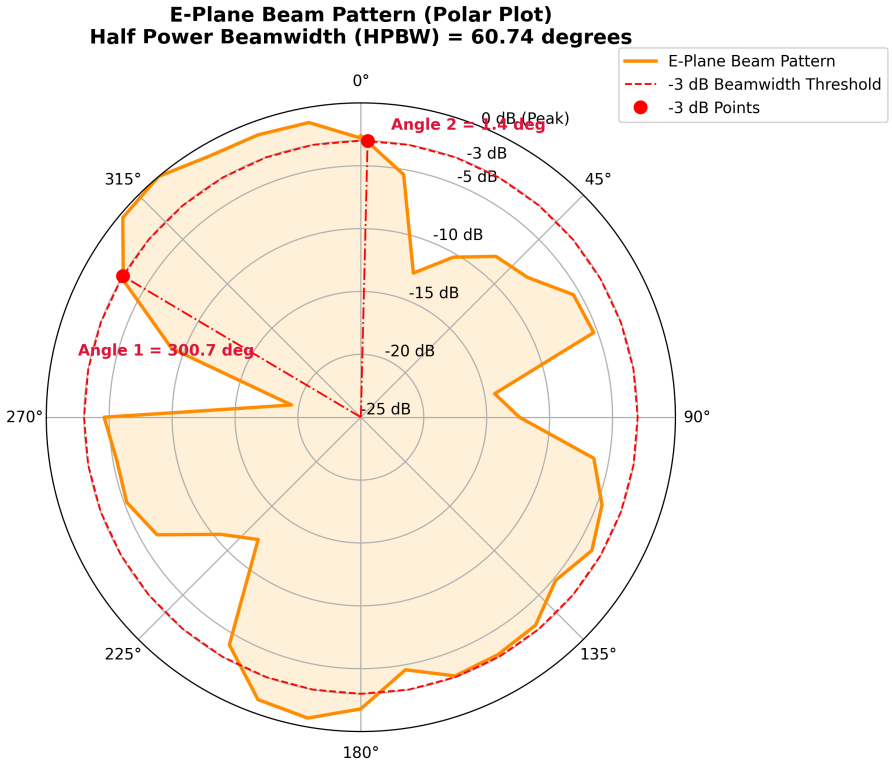


Figure 9: E-Plane Co-Polarization Pattern at $R = 420$ cm in Polar coordinates ($\Delta\theta_E = 60.74°$).

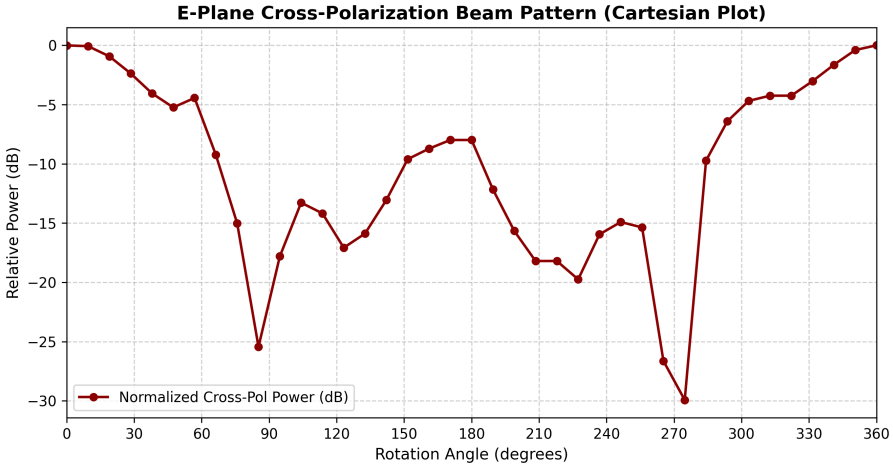


Figure 10: E-Plane Cross-Polarization Beam Pattern at $R = 420$ cm in Cartesian coordinates.

Q2. What criteria must be satisfied to ensure far-field (Fraunhofer) measurement conditions?

The separation distance R between the antennas must satisfy:

$$R \geq \frac{2D_a^2}{\lambda}, \quad R \gg D_a, \quad R \gg \lambda \tag{24}$$

where D_a is the largest physical dimension of the antenna aperture and λ is the operating wavelength. At $R = 420$ cm $\approx 20\lambda$ (where $\lambda \approx 21.13$ cm), the setup operates well inside the far-field region.

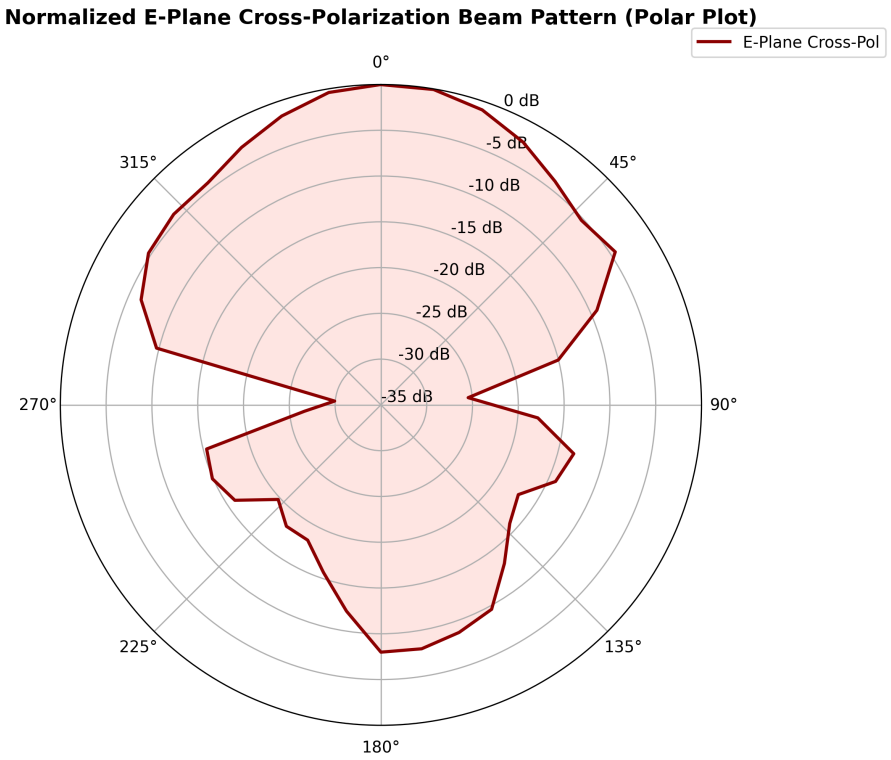


Figure 11: E-Plane Cross-Polarization Beam Pattern at $R = 420$ cm in Polar coordinates.

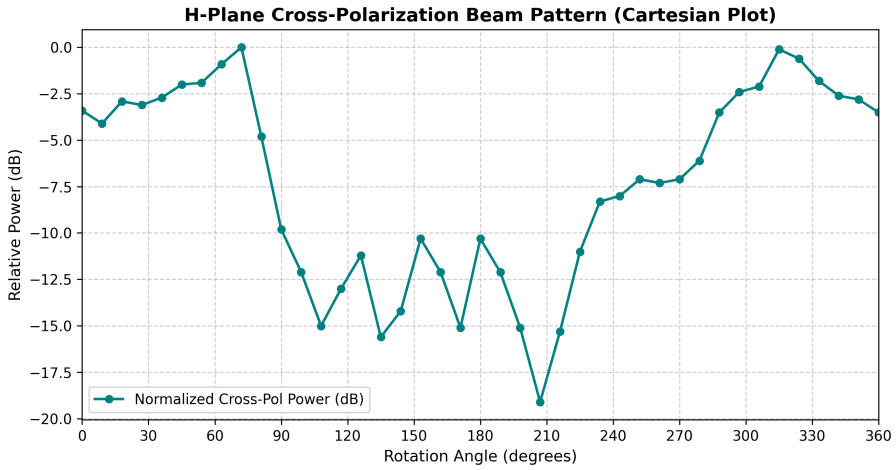


Figure 12: H-Plane Cross-Polarization Beam Pattern at $R = 420$ cm in Cartesian coordinates.

Q3. Why is -3 dB beamwidth evaluation not standard for cross-polarization patterns?

The -3 dB beamwidth (HPBW) characterizes the angular width of an intentional radiating main beam. In cross-polarization, the transmitter and receiver electric fields are placed at 90°, meaning the received signal represents residual leakage and polarization mismatch rather than an intended main lobe. Cross-polarization data is instead evaluated for cross-polarization discrimination (XPD).

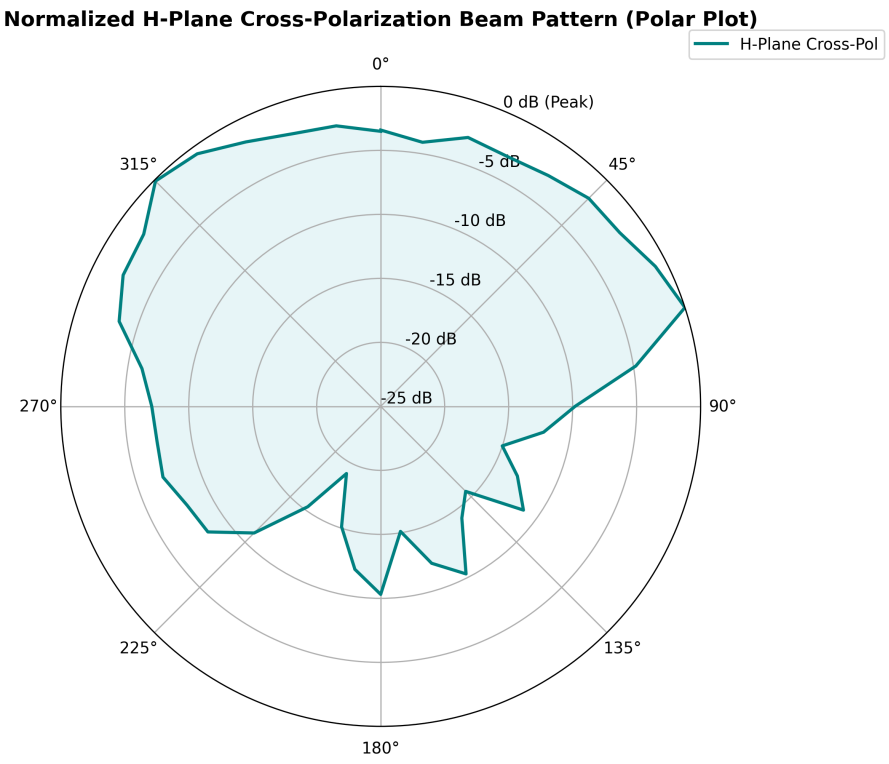


Figure 13: H-Plane Cross-Polarization Beam Pattern at $R = 420$ cm in Polar coordinates.